

JORGE BURGOS

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OBJECTIVE

Computer Engineering student at UPRM with production-level experience building data pipelines, REST APIs, and cloud-hosted database systems using Python, SQL, and PostgreSQL. Experienced with Docker for local testing and data quality validation, ETL pipeline design, and technical documentation. Seeking a Data Engineering internship to apply pipeline and infrastructure skills to high-throughput data systems.

EDUCATION

B.S. in Computer Engineering

Expected May 2027

University of Puerto Rico at Mayaguez — Mayaguez, PR

- Coursework: Database Systems (ICOM 5016), Operating Systems (ICOM 5007), Data Structures (ICOM 4035)

High School Diploma — Industrial Electronics Certification

2020

Benjamin Harrison Vocational High School — Cayey, PR

- Graduated with Academic Excellence

PROJECTS

Car Tracking System — Data Pipeline and REST API

Jan 2026 – Present

CIIC 4060 / ICOM 5016 – Database Systems | UPRM

- Built a Python ETL pipeline using pandas and psycpg2 to ingest, clean, and transform large-scale Parquet datasets into a cloud-hosted PostgreSQL 17 database, applying data quality checks and validation throughout to ensure integrity at every stage.
- Used Docker to containerize the environment for local testing and data manipulation before deploying changes to the production database on Render — preventing unvalidated data from reaching the live system.
- Developed a 28-endpoint REST API using FastAPI to give downstream systems efficient, queryable access to processed data; applied schema normalization and indexing for optimized retrieval performance.
- Documented data schemas, pipeline stages, and API specifications to support team understanding and maintainability; used Git for version control across the project.

Electric Vehicle Reservation System

2025

ICOM 4035 – Data Structures | UPRM

- Designed a Java backend system applying core data structures (HashSet, Queue, LinkedList, Stack) for efficient, constant-time data access and processing across multiple concurrent users.

SKILLS

Languages: Python, SQL, Java, C++, C

Data Engineering: ETL pipelines, pandas, NumPy, data quality validation, data cleaning

Databases: PostgreSQL 17, PostGIS, schema design, query optimization, time-series data handling

Infrastructure: Docker, Git, Linux, cloud deployment (Render)

Backend & APIs: FastAPI, REST API design, psycpg2, handler/DAO architecture

Practices: Technical documentation, integration testing, Agile collaboration

Languages: Spanish (Native), English (Proficient), French (Basic)

WORK EXPERIENCE

Independent Seller

Feb 2020 – Apr 2020

Vector Marketing Corporation — Cayey, PR

- Achieved independent sales through video calling strategies; developed communication and client relationship skills in a remote environment.